

VME crate modules

Starting from the left (slot 1) in the VME64x VXS crate:

1) v6100 single board computer

- currently assigned this name/IP
bbspare1.jlab.org A 129.57.192.55
- debug port can be accessed using
telnet hatsv12 2005

2) FlexIO board w/ two submodules

- Top half handles the FADC internally generated triggers for digitized waveform readout.
 - associated trigger labelled 'DT' on convertor board ("Data Trigger")
- Bottom half handles the Helicity Flip generated triggers for readout of the FADC software scalers, v792 conventional QDC, and conventional scaler
 - associated trigger labelled 'HT' on convertor board ("Helicity Trigger")

3) Level convertor board

- I think this thing is FPGA-based too, but right now it is basically a level convertor translating the high-density PECL signals (small ribbon cable at top) to ECL and NIM signals. Description as follows, starting at top and working down:
 - small PECL ribbon cable to FADC
 - N unused PECL sockets
 - 6 pair ECL socket w/ three labelled pairs
 - 'HF' (INPUT) takes the !MPS signal indicating a helicity blanking interval
 - 'H' (INPUT) takes the Helicity state indicator (HEL)
 - 'DT' (OUTPUT) runs to the upper ECL trig input on the upper FlexIO sub-unit
 - should be copy of the FADC internally generated trigger
 - unused LEMO socket
 - 'HF' (INPUT) NIM equivalent to the above ECL input (use this /or/ ECL input)
 - 'H' (INPUT) NIM equivalent to the above ECL input (use this /or/ ECL input)
 - unused LEMO socket
 - 8 pair ECL socket w/ two labelled pairs
 - 'T0' (OUTPUT) NIM output timed off of HF signal (top pair)
 - behaves like L1A, can be used to gate conventional ADCs etc for readout during MPS blanking interval
 - 'HT' (OUTPUT) runs to the upper ECL trig input on the lower FlexIO sub-unit
 - timed off of the !MPS leading edge for Helicity Trigger

4) Blanked off slot

- NOTE: needed for proper cooling of FADC!

5) JLab F250 FADC board

- 16 LEMO inputs, AC coupled
 - board programmed for input range from 0 -> -2V signal (confirm?)
- 'C' Calorimeter, 'S' Scintillator, 'L' Left, 'R' Right
- NOTE: It's important that the right signal is plugged into the correct input since the FADC will sum and build triggers from specific combinations of the inputs.

6) Blanked off slot

- NOTE: needed for proper cooling of FADC!

7) Blanked off slot

- NOTE: needed for proper cooling of FADC!

8) Conventional scaler (CAEN v560)

- NOTE: Having something two slots over from the FADC board is recommended for proper airflow.

9) CAEN v792 conventional ADC
